COMP 3220 Fall 2023
Assignment 1

Due: 11:59 pm, Sep. 30, 2023

1. We studied the "Process Sale" use case. We are about to study object-oriented analysis and design based on this user case. Pretend you only know user cases and you do not know anything about object-oriented analysis and design yet, but you have to implement the "Process Sale" in Java now. Based on your understanding of this case, implement this use case in any way you like(if you have to make assumptions when implementing, please state it as comments in your code). No GUI is needed and it can be a text-based interaction program. The purpose of this question is not to judge how good/bad your implementation is but to revisit your implementation later after we finish studying OOAD.

2. When identifying use cases, we find a recurring pattern. For any type of object—like inventory items (e.g., a guitar), a record in a database table, and so on—we will usually have to create an item, retrieve an item, update an item, and delete an item. These are called the CRUD operations. There are three options for writing use cases in this situation:
    a. Write a separate use case for each operation.
    b. Write one use case for all of the operations (a CRUD use case).
    c. Omit the use cases from the specification.

   Which of the choices do you think is most appropriate in general? Why?

3. Consider that you are analyzing requirements for a housing system run by the campus housing service.

   The campus housing service helps students find apartments. Apartment owners complete information forms about the available rental units (e.g., location, number of bedrooms, monthly rent), which are then entered into a database.

   Students can search this database via the Web to find apartments that meet their needs (e.g., a two-bedroom apartment for $400 or less per month within a half mile of campus) and contact the apartment owners directly to see the apartment and possibly rent it. Apartment owners call the service to delete their listings when they have rented their apartment(s).

   a) Identify the actors and major use cases for the above high-level business processes in a housing system run by the campus housing service.

   b) Draw a use case diagram for the system, considering all of the actors and use cases you identified in Step 1.

4. Create a set of use cases for the following streaming service (offer a wide variety of TV shows, movies, dramas, and documentaries for their members) membership system.

Members join the system by registering online (username (email), password, address, phone no) and including a credit card to pay the membership fees. Most memberships are for one year, but memberships as short as two months are available.

Throughout the year, the streaming service offers various discounts on its regular membership prices (e.g., two memberships for the price of one for Christmas). It is common for members to pay different amounts for the same length of membership. The service sends out reminder emails to members asking them to renew their memberships one month before their memberships expire.

Some members have become angry when asked to renew at a much higher rate than their original membership contract. That is why the organization wants to track the price paid so that the manager can override the regular prices with special prices when members are asked to renew.

The system must track these new prices so that renewals can be processed accurately. The manager wants the system to track each time a member visits the site. The system will then identify the heavy users and generate a report so that the manager can ask them to renew their memberships early, perhaps offering them a reduced rate for early renewal. Likewise, the system should identify members who have not visited the site in more than a month so that the manager can email them and attempt to re-interest them in the service.

Perform the following tasks:

a) Identify and list actors and use cases. For example, here is one use case for you in advance: (1) maintain service usage records. You should identify and list all other relevant use cases. For the "maintain service usage records" use case, create a brief use case description.

b) Now, write an expanded (in fully-dressed format) use case description for the "maintain service usage records" use case. While doing this, be sure to account for the possibility of the alternative scenario(s) (if there is any).

c) Draw a use case diagram for the system, considering all of the actors and use cases you identified in Step 1 with relationships ("extends", "include") if exists.

Use StarUML or other software of your choice to draw use case diagrams and submit your use case diagrams as pictures with the written use case. Please note that you will not get any credit for hand-drawn diagrams.